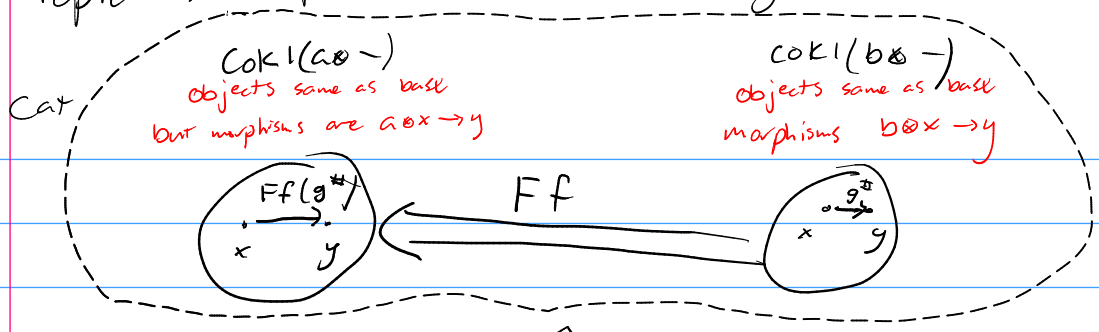
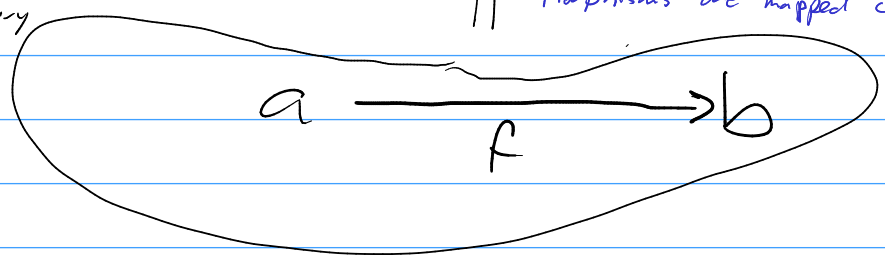
Topic: Are parametric Markov categories indexed categories?



$F : c \mapsto \mathrm{Cokl}(c \otimes -)$ } object mapping

Morphisms are mapped contravariantly to functors.
(are they actual functors?)
Those functors map Cokl maps to other cokl maps

Base category is Markov



How does composition in $\mathrm{Cokl}(c \otimes -)$ work?

$c \otimes -$ is a functor. Objects in $\mathrm{Cokl}(c\otimes -)$ are the same as objects in Base, but a morphism $g^\# : x \to y$ in $\mathrm{Cokl}(c\otimes -)$ corresponds to a morphism $g : c\otimes x \to y$ in Base

$$\begin{tikzcd}
c\otimes c\otimes x \arrow[dr, "id_c \otimes g"] & & c\otimes c\otimes y \\
c\otimes x \arrow[u, "k_x := \Delta_c \otimes id_x"] \arrow[d, "\varepsilon_x := \tau_x = \delta_c \otimes id_x"'] \arrow[rr, bend left, "\hat{g}"] \arrow[drr, "g"] & & c\otimes y \arrow[dr, "h"] \\
x & & y & & z
\end{tikzcd}$$

Identity: $id_x^\# := \varepsilon_x$

Composition: $g^\# \underbrace{\fatsemi}_{\text{Cokl composition}} h^\# := \hat{g} \underbrace{\fatsemi}_{\text{base category composition}} h$

where $\hat{g} := k_x \fatsemi (id_c \otimes g)$ note: I'm leaving out associators here

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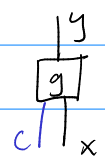
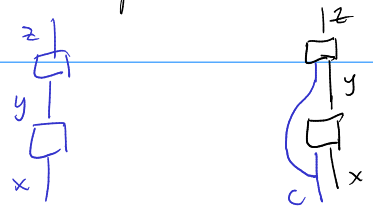
In other words: Base is black with a blue object c
$\mathrm{Cokl}(c\otimes -)$ is blue

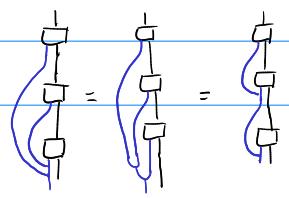
in $\mathrm{Cokl}(c\otimes -)$ =



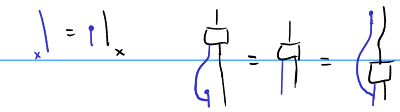
in Base

Composition





Identity



How does Ff map cokleisli maps?

$f : a \to b$ in base

Ff maps objects trivally

For morphisms:

$g^\# : x \to y$ in $\mathrm{Cokl}(b \otimes -)$

ie $g : b \otimes x \to y$ in Base

$(Ff)(g^\#) : x \to y$ in $\mathrm{Cokl}(a \otimes -)$

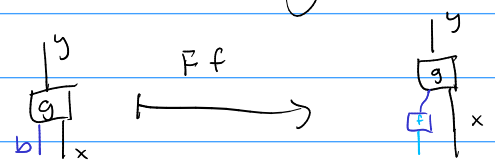
$: a \otimes x \to y$ in Base

in Base

$$a \otimes x \xrightarrow{f \otimes id_x} b \otimes x \xrightarrow{g} y$$

$$Ff(g^\#) = f \otimes id_x \,;\, g$$

In diagram:



Two questions:

1. For $f: a \to b$ in Base, is Ff functorial?

ie

- for $x \xrightarrow{g^\#} y$ and $y \xrightarrow{h^\#} z$ in $\mathrm{Cokl}(b \otimes -)$

  Does $[Ff](\underbrace{g^\# \rightrightarrows h^\#}_{\text{composition in } \mathrm{Cokl}(b\otimes -)}) \overset{?}{=} \underbrace{[Ff](g^\#) \rightrightarrows [Ff](h^\#)}_{\text{composition in } \mathrm{Cokl}(a\otimes -)}$

- Does $[Ff](id_x) \overset{?}{=} id_{[Ff]x}$

2. Is $F: \mathrm{Base} \to \mathrm{Cat}$ functorial?

ie.

- For morphisms $a \xrightarrow{f} b$ and $b \xrightarrow{g} c$ in base,

  Does $F(f \circ g) \overset{?}{=} Fg \underset{\text{functor composition in Cat}}{\circ} Ff$

- Does $F(id_a) \overset{?}{=} id_{\mathrm{Cokl}(a\otimes -)}$

$g^\# : x \to y \qquad h^\# : y \to z$

$g : b \otimes x \to y \qquad h : b \otimes y \to z$

$f : a \to b$

1. $[Ff](g^\# \Rrightarrow h^\#)$

$g^\# \Rrightarrow h^\#$ in $\mathrm{CoKl}(b \otimes -)$

is = to $\hat{g} \mathbin{;} h$ in Base

$= b \otimes x \xrightarrow{\Delta_b \otimes id_x} b \otimes b \otimes x \xrightarrow{id_y \otimes g} b \otimes y \xrightarrow{h} z$

$[Ff](g^\# \Rrightarrow h^\#)$

$= a \otimes x \xrightarrow{f \otimes id_x} b \otimes x \xrightarrow{\Delta_b \otimes id_x} b \otimes b \otimes x \xrightarrow{id_y \otimes g} b \otimes y \xrightarrow{h} z$

Much easier if you know CoKl Composition in string diagram form from the start